

What we're doing

Instead of drawing N contacts-v1 rollouts i.i.d., first compute contact probabilities with the pairwise $P(\text{contact})$ readout, take the top N pairs, and start rollout r by prompting the model with pair r .

Score both ways on eval-val (97 natural FoldBench monomers, #245) with the decontaminated #232 m2-p06 checkpoint: consensus R-precision as usual, and the oracle best rollout per structure as a headroom diagnostic.

Why we expected a tie, and a diversity win

Conditioning on true partial structure is enormously informative: #163 lifted R-precision from 0.145 to 0.556 by prompting with ground-truth partial contacts. The catch has always been that there is no oracle at inference time.

The pairwise readout is a weak stand-in for one — under exp82 the same weights score about 0.086 lower in R-precision under pairwise than under rollout voting.

So consensus was predicted to be a wash: a wrong seed poisons a whole rollout. Oracle best-of-N was predicted to improve, because forcing N distinct starting pairs decorrelates the rollouts, and a broader search is what best-of-N rewards.

Results

Gate first: the unseeded control reproduces #245's published m2-p06 eval-val number, 0.5217 against 0.520, so these arms sit on the same axis as every other MarinFold contact number.

Consensus R-precision on eval-val (n=97), all-range, against a 0.5217 control.

Seeded with the top 100 pairs overall — 0.5234, paired gain +0.0017.

Seeded with 100 long-range pairs only — 0.5244, paired gain +0.0028.

Seeded with the best 33 in each separation range — 0.5247, paired gain +0.0030.

Every one is a tie: all three intervals straddle zero and sit inside #204's 0.005 noise floor. Both preregistered predictions held.

The only significant effect anywhere is oracle best-of-100 under top-100 seeding, +0.0142 [+0.0055, +0.0247] — and that is the arm that does not concentrate its seeds. Restricting all 100 seeds to long range gives the gain up.

Biasing toward long range does not work

The top 100 pairs overall are already 56.8% long-range, because long-separation pairs dominate the candidate universe. Equal thirds therefore lowers the long-range share to 34% rather than raising it.

Neither helps, and the arm aimed at long range is behind the unaimed one at long range: 0.5068 against 0.5081.

Long-range seeds are not intrinsically worse. Inside the equal-thirds arm they are the most accurate at 63.5%, against 49.0% medium and 46.0% short. Depth into the pairwise ranking is what costs a seed its accuracy, not separation range.

Why every arm ties: one contact is almost no conditioning

The rollout index is the pairwise rank of its seed, so the run contains its own dose-response curve, and the curve is flat.

Seeds ranked 1-10 are true contacts 79.2% of the time and the rollouts they produced score 0.3931. Seeds ranked 71-100 are true only 46.0% of the time and their rollouts score 0.3890. Rollouts given no seed at all score 0.3868.

Within a protein a true seed beats a false one by +0.0124 [+0.0082, +0.0168], and a false seed costs nothing against no seed at all. Against #163's 0.145 to 0.556 from conditioning on true partial contact sets, the signal is in the joint structure of many contacts, not in an anchor.

Beware the pooled true-versus-false split, which reads +0.18 and is protein difficulty, not conditioning: good proteins yield both more correct seeds and better rollouts.

The bottleneck is ranking, not diversity

The 100 rollouts already propose 92% of the true contacts, from only 15.7x R distinct pairs, with a mean pairwise Jaccard of 0.257 between any two of them. All three seeded arms match that to within 0.002.

Vote rank recovers 0.52 of the true contacts at the R cut, 0.67 at 2R, 0.79 at 5R and 0.92 at the union. The 0.52 to 0.92 gap is ranking loss, and it is five times larger than anything more diversity could buy.

Re-ranking the pooled pairs does not close it. Cross-validated over proteins, the best R-precision from any combination of vote count, pairwise probability, emission rank and rollout self-consistency is +0.0015 — a tie. Fitting over all candidates instead buys +0.020 AUC and costs 0.0065 R: you can have one or the other, because 99.9% of candidates are pairs no rollout ever proposed.

What to do with it

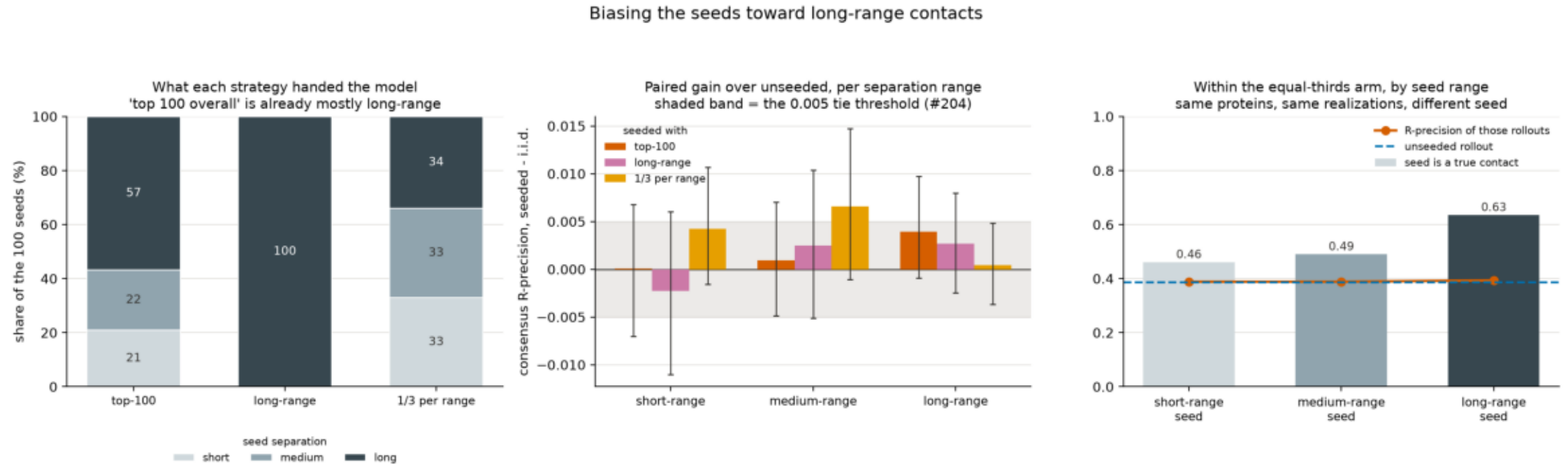
Seeding is not a decoding win, and neither is biasing where the seeds come from. The one thing it bought was diversity, and the rollouts are already diverse.

Pointwise re-ranking is exhausted too — votes, pairwise probability, emission rank and self-consistency all measure the same marginal confidence and agree.

Reaching the 0.92 the sampler already proposes needs something that scores sets rather than pairs: geometric realizability (#211), a folding model in the loop, or joint conditioning on partial maps of size k greater than one, which is where #163 says the signal is. The downstream half — how many of these pairs a folding model actually wants — is Marifold #256.

seed_strategy_eval_val.png

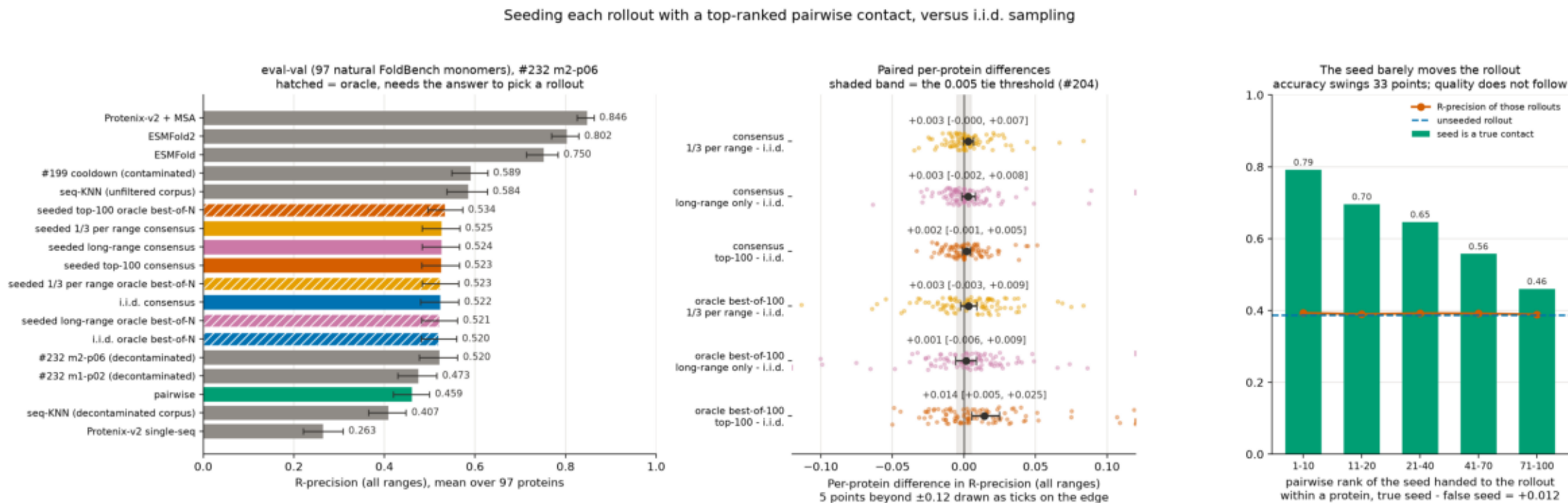
Seed composition per strategy, consensus R-precision per separation range per strategy, and seed accuracy against rollout quality by the seed's own separation range, on eval-val for #232 m2-p06.



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$ plot_strategies.py --data data --out plots
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seeded_vs_iid_eval_val.png

All-range contact R-precision on eval-val for the #232 m2-p06 checkpoint under five MarInFold readouts and #245's published predictors, the paired seeded-minus-i.i.d. differences, and seed accuracy against rollout quality along the pairwise ranking.



\$ plot.py --data data --out plots